INTERACTIVE END-OF-CHAPTER EXERCISES

LONGEST PREFIX MATCHING

Consider a datagram network using 8-bit host addresses.
Suppose a router uses longest-prefix matching, and has the following forwarding table:

Prefix Match	Interface
11	1
10	2
111	3
100	4
011	5
otherwise	6

QUESTION LIST

1. Suppose a datagram arrives at the router, with destination address 01111100. To which interface will this datagram be forwarded using longest-prefix matching?

2. Suppose a datagram arrives at the router, with destination address 01110100. To which interface will this datagram be forwarded using longest-prefix matching?

3. Suppose a datagram arrives at the router, with destination address 00010101. To which interface will this datagram be forwarded using longest-prefix matching?

SOLUTION

1. Since the address is 01111100, it will go to interface 5.

2. Since the address is 01110100, it will go to interface 5.

3. Since the address is 00010101, it will go to interface 6.

QUESTION 1 OF 3

Suppose a datagram arrives at the router, with destination address 01111100. To which interface will this datagram be forwarded using longest-prefix matching?

Answer

CHECK HINT SKIP

We gratefully acknowledge the programming and problem design work of John Broderick (UMass '21), which has really helped to substantially improve this site.

Copyright © 2010-2022 J.F. Kurose, K.W. Ross
Comments welcome and appreciated: kurose@cs.umass.edu